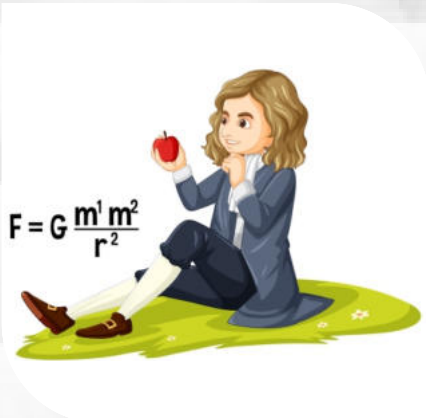
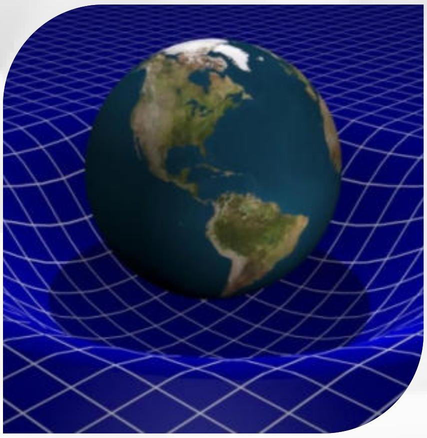
UNIT 06

GRAVITATION

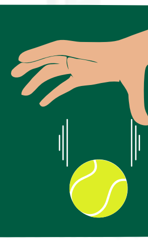




INTRODUCTION OF GRAVITATION

DEFINITION

Gravitation refers to the natural force of attraction existing between all objects in the universe. This force, known as gravity, is responsible for keeping planets in orbit around stars and stars in orbit around the center of galaxies.



EXAMPLE

Imagine two objects, such as a basketball and a bowling ball, placed near each other. Due to the force of gravity, the basketball and the bowling ball are both drawn towards each other, as if they were being pulled together.

EXPLANATION

Gravitation is the force of attraction between all objects in the universe. The force of gravity between two objects depends on their masses and the distance between them. The larger the mass of an object, the greater its gravitational pull. The farther apart two objects are, the weaker the gravitational pull between them.

THE DISCOVERY OF GRAVITY

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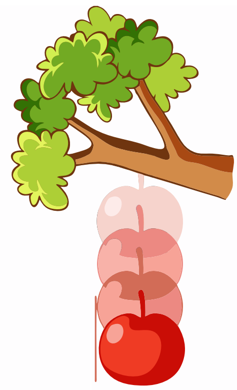
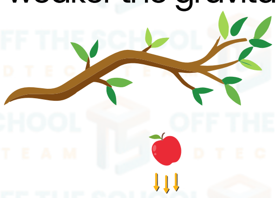
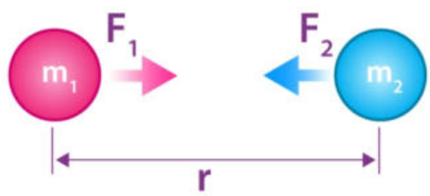




NEWTON'S LAW OF GRAVITATION

Newton's Law of Gravitation is one of the most crucial laws of physics that explains how two objects are attracted to each other. This law was discovered by Sir Isaac Newton in 1687 and explained the behavior of all objects in space, including planets and stars.



STATEMENT

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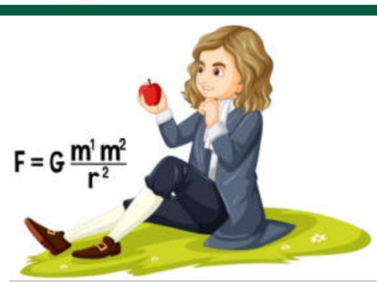


MATHEMATICAL REPRESENTATION

Gravitation is the force of attraction between all objects in the universe. The force of gravity between two objects depends on their masses and the distance between them. The larger the mass of an object, the greater its gravitational pull. The farther apart two objects are, the weaker the gravitational pull between them.

$$F \propto m_1 m_2 \quad \text{\_\_\_\_\_\_\_\_ (i)}$$

$$F \propto \frac{1}{r^2} \quad \text{\_\_\_\_\_\_\_\_ (ii)}$$

Combining equation (i) and (ii);

$$F \propto \frac{m_1 m_2}{r^2}$$

$$F = G\frac{m_1 m_2}{r^2}$$

THE CONSTANT "G"

The constant **"G"** in Newton's Law of Gravitation is known as the gravitational constant. It is a number that represents the strength of the force of gravity between two objects. The value of G has been measured and determined through experiments, and its value is constant for all objects in the universe.

The value of G is approximately **$6.67\times10^{-11}\ Nm^2/kg^2$**.

According to Newton's law of gravitation every two objects attract each other with equal force but opposite in direction.

M_1 = Mass of body A

M_2 = Mass of body B

F_{12} = Force with which body A attracts body B

F_{21} = Force with which body B attracts body A

Then according to the Newton's law of motion;

"For every action, there is an equal and opposite reaction"

$$F_{12} = -F_{21}$$

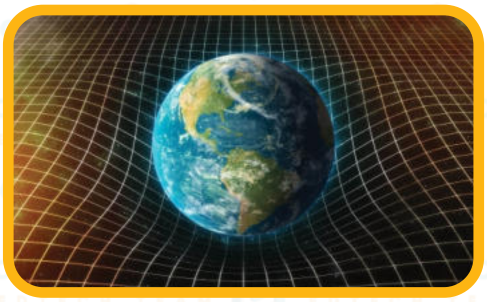
This shows that, the two forces are equal in magnitude but opposite in direction. If F12 is considered as "Action Force" and F21 as "Reaction Force".

FOR EXAMPLE

If you push on an object, it will push back with the same amount of force. If you jump off a diving board, the diving board will push back on you. The important thing to remember is that the force you apply on an object is equal to the force that object will apply back on you Hence, Newton law of gravitation is consistent with Newton's third law of motion.

GRAVITATIONAL FIELD

Gravitational field strength 'g' is the gravitational force acting per unit mass. Any object near the Earth experience this force which is due to gravity of the Earth. The direction of gravitational field strength is directed towards the center of the Earth. The gravitational field strength 'g' is approximately 10 Newton per kilogram 10 Nkg^{-1}. The gravitational field strength 'g' is different at different planets.



FOR EXAMPLE

The gravitational field strength ‘g’ on the surface of Moon is approximately 1.6 Nkg^{-1}.

ACCELERATION DUE TO GRAVITY ‘G’

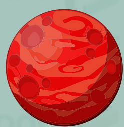
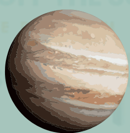
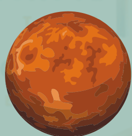
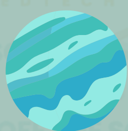
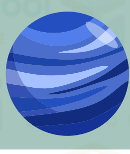
The “g” (acceleration due to gravity) is the acceleration that an object experiences when it falls freely under the influence of gravity.

MOTION UNDER THE INFLUENCE OF GRAVITY

When an object is in free fall, it falls vertically downwards under the influence of gravity and also experiences a constant acceleration of g.

VARIATION IN ‘G’

The gravitational pull an object experiences varies from planet to planet.

SOLAR SYSTEM		VALUE OF g MS^{-2}
Earth		9.8
Moon		1.62
Venus		8.87
Mars		3.77
Jupiter		25.95
Sun		274
Mercury		3.59
Saturn		11.08
Uranus		10.67
Neptune		14.07

Remember that the lowercase "g" and capital "G" are separate, even if they seemidentical. The lowercase "g" is the constant of the gravitational field and is usually used to represent the gravity of Earth. The uppercase "G" is the constant of gravity between two bodies.

Gravitational Constant "G"	Acceleration of Gravity "g"
• The gravitational constant "G" is a numerical value that determines the strength of the gravitational force between two objects.	• "g" is the acceleration due to gravity, it represents the rate at which objects fall towards the ground when dropped.
• The unit of "G" is Nm^2/kg^2.	• The unit of "g" is m/s^2.
• The purpose of "G" is to calculate the force of attraction between two objects due to gravity.	• The purpose of "g" is to calculate the acceleration of an object that is falling freely under the influence of gravity.
• "G" is a constant that does not change with location and altitude.	• "g" changes with location and altitude.

WEIGHT

DEFINITION

Weight is the force with which a body is attracted towards the center of the earth by the earth's gravitational field.

MATHEMATICALLY,

Weight of an object of mass 'm' in a gravitational field of strength 'g' is given by the relation: ***W = mg***

Where W is the weight, m is the mass of the object, and g is the acceleration due to gravity. Weight is commonly measured in Newton's (N).

MASS OF THE EARTH

The mass of the Earth can't be measured directly by placing it on any weighing scale. But it can be measured by an indirect method. This method utilizes the Newton's law of universal gravitation.

To calculate the mass of the Earth, let us consider

M_b → Mass of the Ball

M_E → Mass of the Earth

G → Universal Gravitational Constant

R_E → Radius of the Earth (i.e. the distance between the ball and the centre of Earth)

According to Newton's Law of Universal Gravitation, the gravitational force of the Earth acting on the ball is:

$$F = \frac{GmM_E}{R_E^2} \_\_\_\_\_\_\_\_\_\_ (i)$$

Whereas the force of the attraction of the Earth is equal to the weight of the ball. Therefore:

$$F = W = mg \_\_\_\_\_\_\_\_\_\_ (ii)$$

Compare equation (i) and (ii); we get:

$$mg = \frac{GmM_E}{R_E^2}$$

Rearranging the equation gives

$$M_E = \frac{gR_E^2}{G} \_\_\_\_\_\_\_\_\_\_ (iii)$$

Values of constants in equation (iii) are:

$g = 10\ Nkg^{-1}$

$R = 6.38 \times 10^6\ m$

$G = 6.67 \times 10^{-11}\ Nm^2\ Kg^{-2}$

Substituting these values in equation (iii)

$$M_E = \frac{10\ NKg^{-1}\ (6.38 \times 10^6 m)^2}{6.673 \times 10^{-11}\ Nm^2 kg^{-2}}$$

$M_E = 6.0 \times 10^{24}\ kg$

Thus, the mass of the Earth is M_E $= 6.0 \times 10^{24}$ kg

SATELLITES

A satellite is an object in space that orbits around a larger body, such as a planet. There are two different types of satellites:

NATURAL SATELLITES

Natural satellites are objects in space that orbit around a planet. They are also known as moons and are not artificial and serve various purposes, such as the moon helping stabilize Earth's rotation and affecting the tides.

ARTIFICIAL SATELLITES

Artificial satellites are man-made objects that are sent into space to orbit around the Earth or other celestial bodies. These satellites are used for a variety of purposes, including communication, navigation, scientific research, and military surveillance.

HOW DO SATELLITES REACH SPACE?

Satellites are launched into space through the use of powerful rockets. The rockets are designed to propel the satellites high up into the atmosphere and break free from the Earth's gravity. To achieve this, they burn a significant amount of fuel, enabling them to reach orbit around the Earth. Once the rocket reaches a specific altitude and speed, the satellite separates from the rocket and begins to orbit the Earth. The satellite is equipped with engines and guidance systems, which enable it to adjust its orbit and maintain its position to perform its intended function.

EXAMPLES

COMMUNICATION

Artificial satellites are used for communication, transmitting phone calls, TV signals, and internet data.

GLOBAL POSITIONING SYSTEM (GPS)

Artificial satellites are also used for navigation, such as GPS **(Global Positioning System)** satellites. They orbit at high altitudes, and GPS receivers on the ground use signals from multiple satellites to calculate precise locations.

EXPLORATION

Artificial satellites are also used for scientific research. Space telescopes like the Hubble Space Telescope are used for observing stars, galaxies, and other celestial objects.

ORBITS

The path that the satellite follows around the Earth is called its orbit. The time to complete its orbit varies as per the altitude and the angle of its rotation.

NEWTON'S LAW OF GRAVITATION IN THE MOTION OF SATELLITES

The law of gravity made by Newton explains how satellites can move in their orbits. This law can also be used to calculate the centripetal force required for a satellite to revolve around the Earth.

CALCULATING THE VELOCITY REQUIRED FOR A SATELLITE TO STAY IN ORBIT

To calculate the velocity of a satellite, Let us consider the motion of a satellite which is revolving around the Earth;

m → Mass of the Satellite
M → Mass of the Earth
R → Radius of the Earth
h → Height (Altitude) Of the Satellite From the Surface of Earth
r = R + h → Radius of Orbit
We know that,
Centripetal Force = Gravitational Force

$$F_C = F_G \quad \text{(i)}$$

$$F_C = \frac{mV^2}{r} \quad \text{(ii)}$$

$$F_G = G\frac{mM_E}{r^2} \quad \text{(iii)}$$

Using in equation (i);

$$\frac{mV^2}{r} = G\frac{mM_E}{r^2}$$

$$V^2 = \frac{GM}{r}$$

$$[r = R+h];$$

$$V^2 = \frac{GM}{R+h}$$

$$v = \sqrt{\frac{GM}{R+h}}$$

This formula gives the velocity of a satellite in orbit of the Earth. Remember, no matter how big or small a satellite is, it will move at the same speed when it orbits around something.

CALCULATING THE TIME IT TAKES FOR A SATELLITE TO COMPLETE ITS ORBIT

To calculate the time required for a satellite to complete a revolution around its orbit, we will need to do the following steps:
T → Time Required by a Satellite to Complete a Revolution around its Orbit
MATHEMATICALLY,

$$T = \frac{2\pi r}{v} \quad \text{(i)}$$

The velocity of satellite is given by equation;

$$v = \sqrt{\frac{GM}{R+h}}$$

By substituting the formula of velocity of satellite in equation **(i)**, we get

$$T = \frac{2\pi r}{\sqrt{\frac{GM}{R+h}}} \qquad \because \ r = R + h$$

$$T = 2\pi r\sqrt{\frac{r}{GM}}$$

$$T = 2\pi\sqrt{\frac{r^3}{GM}}$$

This equation can be used to calculate the time it takes for a satellite to complete its orbit.

MOTION OF ARTIFICIAL SATELLITE AROUND THE EARTH

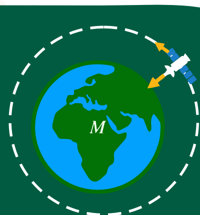
To make satellites revolve around Earth, the speed at which they travel is carefully calculated; so they do not fall back to Earth. Newton explained this phenomenon through an experiment.

NEWTON'S EXPERIMENT

PATH A	PATH B	PATH C
The ball shot from the cannon falls down after a while and does not reach orbit.	The ball shot from the cannon falls down after a while and does not reach orbit.	The ball shot from the cannon does not fall down after a while and escapes the gravity of Earth but gets caught up in its orbit, so it starts revolving around the Earth.
This happens due to the difference in the launch speeds of the balls.		
The launch speed of the cannonball is slow.	The launch speed of the cannonball is medium.	The launch speed of the cannonball is fast enough to escape Earth's gravity.

ORBITAL VELOCITY

"The velocity required to keep the satellite in orbit is called Orbital Velocity."



CALCULATING THE ORBIT VELOCITY

To calculate the velocity required for a satellite to stay in orbit,
we will need to do the following steps:
First we calculate the weight of the satellite,
$W_s = mg$
$W_s \rightarrow$ Weight of the Satellite, therefore
$F_c = W_s$ and $W_s = mgh \rightarrow$ (I)
Where,
M $\rightarrow$ Mass of the Satellite
$g_h \rightarrow$ Acceleration Due to Gravity at Height "h" From the Surface of Earth
The centripetal force "F" acting on the satellite is:

$$F_c = \frac{mv^2}{r}$$

Substituting the values of "F" and "W" in equation **(i)**:

$$\frac{mv^2}{r} = mg_h$$

$$v^2 = g_h r$$

$$v = \sqrt{g_h r}$$

$$r = R + h$$

$$v = \sqrt{g_h (R + h)}$$

If the satellite is orbiting very close to the surface of the Earth
then: h << R

In this case orbital radius may be considered equal to radius of Earth
Therefore,

$$R + h = R$$

Also $g_h = g$

$$V = V_C$$

Where,
$V_c \rightarrow$ Critical Velocity
G $\rightarrow$ Acceleration due to gravity on the surface of the Earth
The equation becomes:

$$V_C = \sqrt{gR}$$

This equation is called **"critical velocity"**.
It can be defined as:
"The constant horizontal velocity required to put the satellite into a stable circular orbit around the Earth."

It is also called proper speed or orbital speed.
If,

$$g = 10 \text{ ms}$$
$$R = 6.38 \times 10^6 \text{ m}$$

Then the orbital equation becomes

$$V_C = \sqrt{gR} = \sqrt{10ms^{-2} \times 6.38 \times 10^6\, m}$$
$$V_C = 7.99 \times 10^3 \text{ ms}^{-1}$$
OR
$$V_C = 8 \text{ kms}^{-1}$$

Remember, the gravity acting on the satellite increases as it gets closer to Earth. So, the satellites in order to stay in an orbit closer to the Earth needs to travel faster compare to those satellites in the farther orbits.

WORKED EXAMPLES

Calculate the speed of a satellite which orbits the Earth at an altitude of 1000 kilometres above Earth's surface?

STEP 1

Write down known quantities and quantities to be found:
$M_{Earth} = M = 6.0 \times 10^{24}$ kg
$R_{Earth} = R = 6.38 \times 10^6$ m
$h = 1000$ km $= 1000 \times 10^3 = 1 \times 10^6$
$G = 6.673 \times 10^{-11}$ Nm kg^{-2}
$v = ?$

STEP 2

Write down formula and rearrange if necessary:

$$v = \sqrt{\frac{GM}{R + h}}$$

STEP 3

Put the values in formula and calculate:

$$v = \sqrt{\frac{6.673 \times 10^{-11}\, Nm^2\, kg^{-2} \times 6.0 \times 10^{24}\, kg}{6.38 \times 10^6\, m + 1 \times 10^6\, m}}$$
$$v = 7.36 \times 10^3 \text{ ms}^{-1}$$
OR
$$v = 8.0 \text{ kms}^{-1}$$

Hence, the orbital speed of satellite is 7.36×10^3 ms^{-1}.